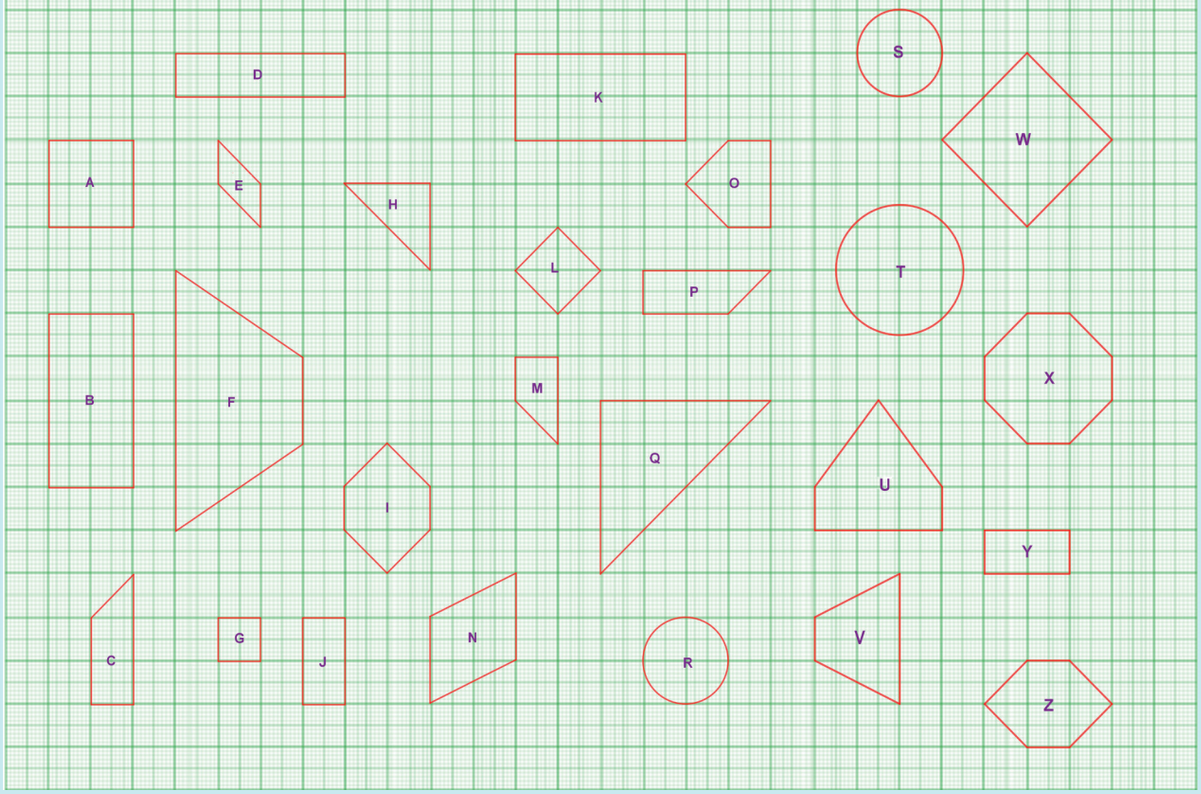
Try these

Identify the pairs of figures which are similar and congruent and write the letter pairs.



5.2.1 Congruent Triangles

Consider two given triangles PQR and ABC. They are said to be congruent (≡) if their corresponding parts are congruent. That is PQ=AB, QR=BC and PR=AC and also $\angle P = \angle A$, $\angle Q = \angle B$ and $\angle R = \angle C$. This is denoted as $\Delta PQR \equiv \Delta ABC$.

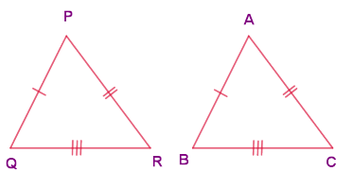


Fig. 5.2

There are 4 ways by which one can prove that two triangles are congruent.

(i) SSS (Side – Side – Side) Congruence

If the three sides of a triangle are congruent to the three sides of another triangle, then the triangles are congruent. That is $AB = PQ$, $BC = QR$ and $AC = PR$

$\Rightarrow \Delta ABC \equiv \Delta PQR$.

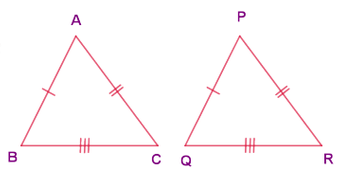


Fig. 5.3

(ii) SAS (Side – Angle – Side) Congruence

If two sides and the included angle (the angle between them) of a triangle are congruent to two sides and the included angle of another triangle, then the triangles are congruent. Here, $AC = PQ$, $\angle A = \angle P$ and $AB = PR$ and hence $\Delta ACB \equiv \Delta PQR$.

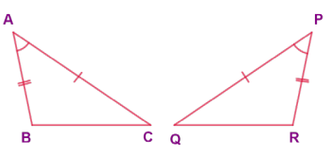


Fig. 5.4

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